

Oluwatobi Melvyn Mayungbo

AI Engineer, LLM and Retrieval Systems

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SUMMARY

AI-native engineer who builds with AI tools and ships working projects. Built an MCP server that lets AI assistants check crypto wallets and tokens for risk across five chains, a retrieval assistant with guardrails, and a payments ledger API. Holds the Claude Code in Action certificate.

SKILLS

AI and LLM Systems: Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), LLM Evaluation and Guardrails, Prompt Engineering, Hybrid Search (BM25, Dense Embeddings, RRF), Citation Verification, Hugging Face, ONNX Runtime

Payments and Risk: Web3 Fraud Risk Scoring, Double-Entry Ledgers and Idempotency, Transaction Fraud Detection, Credit Risk Decisioning, Payment Reconciliation, Regulatory Compliance (CBN/NDPA), Demand Forecasting

Engineering and MLOps: Python, FastAPI, PostgreSQL, React, Docker, CI/CD (GitHub Actions), SQL, DuckDB, Redis, GCP Cloud Run, AWS ECS Fargate, Railway, Prefect, dbt-core, MLflow, Evidently, Git, Pytest, Streamlit

Machine Learning: PyTorch, Scikit-learn, LightGBM, XGBoost, Deep Learning, Sequence Models (GRU, TCN, Transformers), Model Calibration (Isotonic Regression, Temperature Scaling), SHAP, Feature Engineering

Data Analysis: Pandas, NumPy, SQL, Anomaly Detection

PROJECTS

web3-risk-mcp: Web3 Fraud Risk MCP Server | [GitHub](#)

Tech: Python, Model Context Protocol (MCP), Etherscan V2, GoPlus, DexScreener, Docker, GitHub Actions

- Built an MCP server that gives AI assistants (Claude Desktop, Cursor) six read-only tools to check a crypto wallet, token or smart contract for risk before anyone interacts with it, returning a 0 to 100 risk score with a reason for every point, across Ethereum, Base, Arbitrum One, Polygon PoS and BNB Chain.
- Detects honeypot tokens, rug pull signs, hidden owner powers (mint, blacklist, pause), buy and sell tax, holder concentration, unlocked liquidity, and fund links to mixers, sanctioned wallets, exploiters and phishing addresses over 1 or 2 hops.
- Designed a rule-based, fully explainable scorer where findings carry stable IDs and decisive findings set a floor of 75 so trust signals can never hide them; each score carries a confidence level based on how many sources answered.
- Built a resilient HTTP layer with caching, per-source rate limits and retries with exponential backoff and jitter; one failing source never breaks an investigation, and missing data lowers confidence rather than the score.

Fintech Compliance RAG Assistant | [GitHub](#) | [Live App](#)

Tech: Python, Streamlit, BM25, ONNX Runtime, Reciprocal Rank Fusion, FastAPI, Docker, GCP Cloud Run

- Built and deployed a regulatory question-answering system over central bank (CBN) circulars and the NDPA, combining sparse (BM25) and dense (ONNX-served MiniLM) retrieval fused via reciprocal rank fusion, reaching 0.96 recall@5 and 0.842 MRR on a 50-question labelled golden set with zero retrieval misses in the top 10.
- Designed generation to fail safely under uncertainty: a sentinel-token refusal mechanism enforced across the prompt, output guardrail and evaluation harness, plus post-generation citation verification that strips any marker the model cited but never retrieved.

Production-Grade Double-Entry Ledger API | [GitHub](#)

Tech: Python 3.12, FastAPI, PostgreSQL 16, Redis 7, Alembic, Docker

- Built a double-entry ledger and payments API where correctness under concurrency, retries, partial failure and replay is enforced in the database: integer minor units, every transaction summing to zero through a deferred constraint trigger, and an append-only ledger where corrections are reversals.
- Wrote 138 tests against real PostgreSQL and Redis, including a concurrency test where exactly half of N simultaneous transfers succeed and no balance goes negative.

FPL AI Manager | [GitHub](#)

Tech: Python, SQLite, FastAPI, React, Docker, Render

- Built two models that play Fantasy Premier League against the live FPL API: a Manager that carries one squad all season under real transfer, budget and chip rules, and a Best XI optimiser that rebuilds the best legal squad each gameweek.
- Wrote a rules engine, an expected-points model over a multi-gameweek horizon and a leakage-free backfill; cached, rate-limited API client, scheduled jobs and one-command deployment, with 574 tests.

EXPERIENCE

Machine Learning Instructor, SQI College of ICT March 2026 to Present

- Teach machine learning and deep learning to student cohorts, covering statistical foundations, classical ML algorithms, neural architectures, sequence models and production evaluation metrics.
- Instruct students on data analysis, feature engineering and model building with PyTorch, Scikit-Learn, Pandas and NumPy.

Internal Auditor, NISER May 2024 to July 2024

- Analysed financial and operational records to identify inconsistencies and irregularities, performing manual anomaly detection across complex transactional datasets.

EDUCATION

B.Sc. Statistics, University of Ibadan January 2021 to March 2025

Professional Diploma in Artificial Intelligence, SQI College of ICT July 2025 to Present

CERTIFICATIONS

- Google Data Analytics Professional Certificate
- Google Advanced Data Analytics Professional Certificate
- Claude Code In Action